Resource-Light Evaluation of Weather-Aware Vegetation Forecasting: Simple Baselines, Frozen Encoders, and Sign Without Rate

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Abstract

On 115 GreenEarthNet minicubes from tile 32UNU, a ridge read-out over frozen Earth-observation embeddings plus weather predicts cube-mean NDVI 5–100 days ahead, clearing a proxy climatology, an observation-process control, and a permutation null while surviving a geographic block holdout. On paired per-fold differences, however, no frozen encoder view—including colour-infrared re-encodes—separably beats a 21-column summary of the same bands at any horizon, and all nine are separably below it at 5 days. A 17-column baseline with current NDVI plus weather, and no image at all, beats every frozen network at 5 and 25 days. A delta probe explains the split: frozen representations recover the *direction* of vegetation change, but not its *magnitude*. No frozen network beats persistence on both extreme tails at any horizon, so pooled R^2 is the wrong operational metric for drought and heat stress. Read as a handshake with EO-WM rather than a contradiction, the result is a ground-up evaluation claim: state and sign are already in the representation, while rate enters through the weather pathway.

1 Introduction

Short-horizon vegetation forecasts matter for climate adaptation because drought and heat impacts are expressed locally in vegetation state before they appear in coarse weather products. Operational services already think in multi-week NDVI anomalies—for example early-drought products at 10/30/90-day leads and Copernicus vegetation-stress indicators—and the decision is often binary at a horizon: *will this window fall below a vegetation-stress trigger at a 25-day lead?* That question is asked by drought monitoring desks, agricultural and forestry analysts, and ecosystem observatories. It is also asked by model builders who must decide whether a large frozen foundation model is worth the compute when a small ridge might already answer it.

EarthNet2021 framed weather-conditioned Earth-surface forecasting as a guided video task [5]; GreenEarthNet sharpened that framing to high-resolution vegetation forecasting from Sentinel-2 and meteorology [2]. The 2026 climate-AI agenda emphasises ground-up, resource-aware evaluation: smaller or hybrid models, open tools, and honest accounting of what signal is actually available. In that setting, the central question is not whether a frozen vision encoder contains *some* signal. It is whether that signal adds climate-relevant skill beyond the baselines a practitioner would deploy first—band statistics, current NDVI, and weather—and whether pooled skill hides failure on the stress tails that matter for adaptation.

We study that question with all encoders frozen and no new architecture. On 115 GreenEarthNet minicubes from tile 32UNU, ridge regression over frozen embeddings plus weather predicts cube-mean NDVI 5–100 days ahead and clears every control we test. Attribution does not follow. No frozen encoder separably beats a band-matched hand-crafted baseline at any horizon, and current NDVI plus weather alone beats every frozen network at 5 and 25 days. Companion probes show that frozen appearance features carry the *sign* of vegetation change but not its *rate*, and that weather explains only a within-season *proxy* fraction of the residual anomaly on this single-year tile. The practical message is evaluative: for short-horizon vegetation monitoring, put modelling effort into the weather pathway and into decision-oriented metrics, rather than assume a larger frozen appearance backbone recovers severity.

2 Related Work

EarthNet2021 and GreenEarthNet established the benchmark setting we inherit [2, 5]. EO-WM makes the weather pathway explicit by decomposing forcing into climatology, anomalies, and cumulative stress [3]. The handshake with our study is conceptual. EO-WM already separates direction-sensitive and amplitude-sensitive diagnostics (DHR vs. DAE), reports a larger relative gain on direction than on amplitude versus Wan2.1, and notes in its Table 5 that classifier-free guidance can raise magnitude reproduction without improving ranking. We observe an analogous split one stage upstream, before sequence modelling: frozen appearance representations preserve the sign of vegetation change more readily than its magnitude. We are not reproducing EO-WM's tables (different masks, they train, we freeze, they predict pixels) and we do not treat their amplification setting as the main result—they already declined to.

On the representation side we probe DINOv2 [4] and Satlas-pretrained remote-sensing encoders [1]. The contribution is evaluative, not architectural: whether large pre-trained features already contain the climate signal of interest more economically than small band-statistics baselines and persistence/climatology controls.

3 Setup

Data and target. We use 115 GreenEarthNet minicubes from tile 32UNU, all from 2018. Each cube is 128×128 pixels at 20 m over an approximately 150-day window [2]. The forecast target is cube-mean NDVI at $\Delta \in \{5, 25, 50, 100\}$ days. Tile 32UNU has no seasonal-split coverage, so the leave-year-out climatology used elsewhere in the benchmark is not computable here; weather ceilings on this tile are therefore within-season *proxies*, not H1.

Frozen predictors and baselines. Predictors are frozen EO/image encoders with a ridge read-out. We compare them to a band-matched hand-crafted summary (`raw_rgb_only`; 21 columns of the same RGB bands the networks receive) and to a 17-column autoregressive baseline $[\mathrm{NDVI}(t), \mathrm{weather}]$ with no image. Full `raw_features` additionally sees all four bands and NDVI statistics; we report it for context but do not treat it as a network. A multi-image Satlas encoder is a positive control and is not single-image comparable (one embedding can pool up to eight retained frames over a 0–105 day lookback).

Protocol. Evaluation is leakage-safe at the cube level (cube, LOCO, and spatial-block splits); effective n is counted in cubes. "X beats Y" uses paired per-fold differences with a delete-one-fold jackknife, not overlapping marginal intervals. A plausibility screen drops forecast rows that touch three cloud-contaminated frames (midsummer cube-mean NDVI below zero at clear fractions 0.59–0.63) out of 1580 retained frames. Controls: persistence, proxy climatology, observation-process, horizon-alone, and a permutation null.

4 Results

4.1 Forecasting works; the representation is not the reason

Table 1 summarises the Tier-1 forecast table. Skill is real: the best rows reach $R^2 = 0.871/0.743/0.625/0.639$ at 5/25/50/100 days. The observation control never exceeds 0.072, the horizon-alone control is negative at every horizon, and the permutation null sits near empirical zero. Encoder rows remain positive under spatial blocking, even though the proxy climatology collapses there.

On paired per-fold differences, *no* frozen encoder view separably beats `raw_rgb_only` at any horizon (Figure 2). At 5 days all nine are separably *below* it (-0.141 to -0.267). Colour-infrared twins move some networks by at most $+0.14$ and never enough to reach the baseline. The 17-column $[\mathrm{NDVI}(t), \mathrm{weather}]$ row beats every frozen network at 5 and 25 days. For a drought desk that needs a short-lead answer, that is the resource-aware default until a representation proves otherwise.

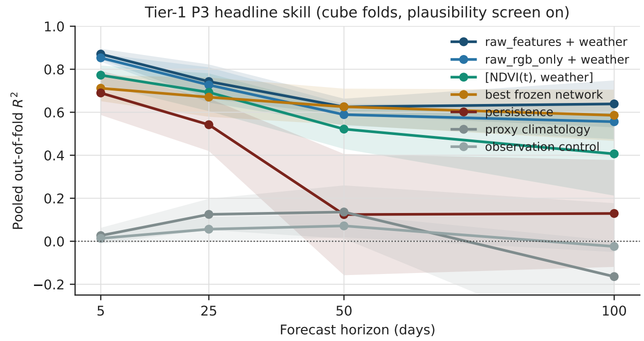

Figure 1: Headline pooled R^2 versus horizon (Tier-1 P3, cube folds, plausibility screen on). Shaded bands are delete-one-fold jackknife intervals. Baselines and weather clear the observation and proxy-climatology controls; the best frozen network stays below the band-matched hand-crafted row.

4.2 Direction without rate; tails without pooled skill

Table 2 and Figure 3 interpret the forecast. Embedding deltas recover sign robustly but not magnitude—the same dissociation EO-WM encodes in DHR versus DAE. Weather alone explains about 0.09–0.13 of the within-season residual anomaly on this tile. A leave-year-out check on seasonal tile 30TVN is inconclusive (proxy $+0.192$ vs. real -0.136; intervals overlap because the crossed fold count clamps to four years), so we do not quote Stage B as H1.

Table 3 and Figure 4 are the climate-facing reading of the same predictions. Pooled R^2 lives in near-normal rows. On the extreme subset, frozen networks are negative against persistence in `extreme_low` at 5–25 days, and no frozen network is positive on *both* extreme tails at any horizon. That is exactly the risk of sign without rate for heat and drought: a model can look strong overall and still miss severity on the windows an early-warning trigger would fire on.

5 Implications and Limits

The practical division of labour is simple. Use small, auditable baselines as the first line for short-horizon vegetation-stress questions; require any frozen encoder—or any world model—to beat $[\mathrm{NDVI}(t), \mathrm{weather}]$ and to report sign and magnitude separately; put modelling capacity into the weather pathway where physically meaningful stress can enter. That is a ground-up evaluation recommendation, not an FM dunk.

Limits are real. This is one Alpine-foreland tile and one year; the weather ceiling is a within-season proxy; Stage B's second-order leak path remains unquantified and is not quoted; P2/P4 are being re-aligned to the same plausibility screen as P3; and we predict cube-mean NDVI, not pixels. These limits argue for restraint and for a follow-on extreme-tile check on the 2018 heat/drought split—not for ignoring the result.

Model row	5 d	25 d	50 d	100 d
raw_features + weather	0.871	0.743	0.625	0.639
raw_rgb_only + weather	0.853	0.727	0.589	0.556
$[\mathrm{NDVI}(t), \mathrm{weather}]$ only	0.773	0.693	0.522	0.406
best frozen network	0.712	0.670	0.626	0.586
persistence	0.690	0.542	0.124	0.129
proxy climatology	0.026	0.125	0.136	−0.164
observation control	0.014	0.056	0.072	−0.024
horizon-only control	−0.006	−0.006	−0.007	−0.023
permutation null	−0.014	−0.002	−0.007	−0.024

Table 1: Headline pooled out-of-fold R^2 for cube-mean NDVI on 115 cubes (tile 32UNU). Effective n in cubes: 115/114/115/94. Nested-CV ridge, shared $[\mathrm{NDVI}(t), \mathrm{weather}]$ base, plausibility screen on. "Best frozen network" is the strongest of nine encoder views per horizon.

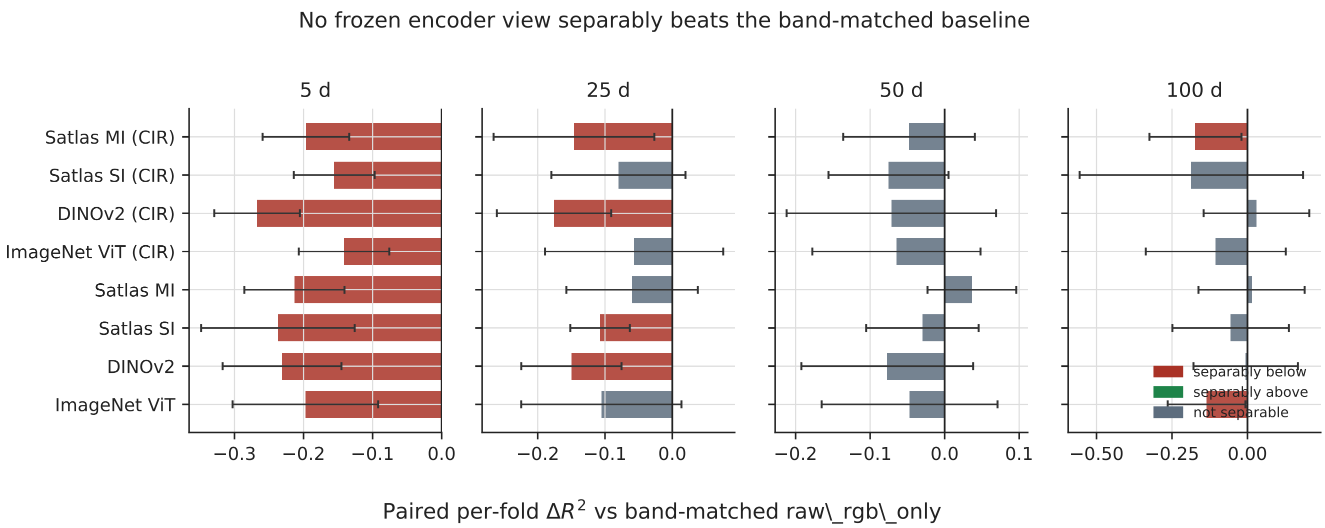

Figure 2: Paired per-fold ΔR^2 of each frozen encoder view against band-matched `raw_rgb_only`. Red bars are separably below; no view is separably above at any horizon.

Probe	Key measured result
P2 sign	DINOv2: $\rho = 0.606$ $[0.546, 0.665]$; margin $+0.54$ over the gap-length control. Band-matched `raw_rgb_only` reaches 0.695.
P2 magnitude	No encoder recovers rate: all $\rho \leq 0.12$; 3 of 4 encoders flip margin sign across fold modes.
P4 proxy ceiling	Weather explains 0.130 $[0.063, 0.197]$ (`cube`) and 0.085 $[0.007, 0.162]$ (LOCO) of the within-season post-*proxy*-climatology anomaly at `cell_mean`/HGB; spatial blocking kills it.

Table 2: Companion probes. Frozen appearance features carry state and direction; rate enters through the weather path. The P4 number is a proxy ceiling, not H1.

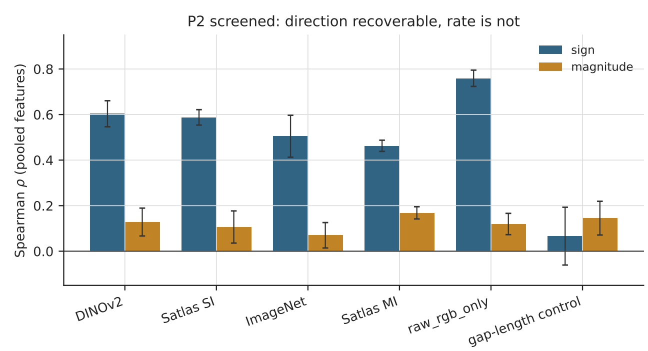

Figure 3: P2 screened cube-mean deltas (pooled features, linear read-out): Spearman ρ for sign versus magnitude. Direction clears the gap-length control; rate does not.

6 Conclusion

A resource-light read-out over Sentinel-2 plus E-OBS already forecasts cube-mean NDVI well above climatology on GreenEarthNet. Frozen EO foundation models are not, in this study, the reason. Direction of change is in the pictures; rate is not; and the extreme tails do not follow the pooled headline. That is useful climate-AI evidence: evaluate what is available before scaling what is fashionable.

References

[1] Favyen Bastani, Piper Wolters, Ritwik Gupta, Joe Ferdinando, and Aniruddha Kembhavi. Satlaspretrain: A large-scale dataset for remote sensing image understanding. In *Proceedings of the IEEE/CVF International Conference*

Skill vs. persistence	5 d	25 d	50 d	100 d
`raw_rgb_only` / extreme_low	+0.12	+0.30	+0.50	+0.69
`raw_rgb_only` / extreme_high	+0.75	+0.38	−0.72	−0.57
DINOv2 / extreme_low	−1.31	−0.51	+0.34	+0.68
DINOv2 / extreme_high	+0.35	+0.29	−0.31	−0.62

Table 3: Decision-oriented diagnostic: skill against persistence in the extreme tails (cube-mean, cube folds, Tier-1 protocol). No *frozen network* is positive on both `extreme_low` and `extreme_high` at any horizon. Hand-crafted rows can clear both tails at short leads and still fail on `extreme_high` at 50–100 d. This is a diagnostic for early warning, not a claim of operational deployment.

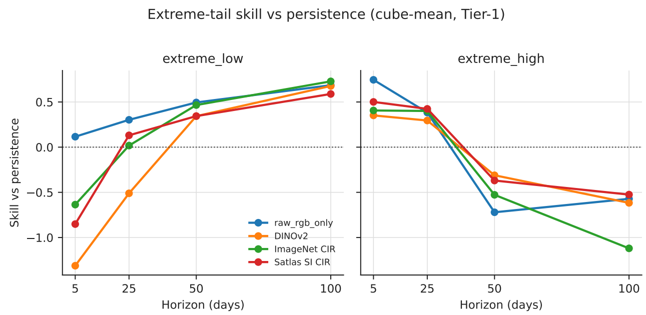

Figure 4: Skill against persistence in the extreme tails (Table 3). No frozen network is positive on both tails at any horizon.

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[2] Vitus Benson, Claire Robin, Christian Requena-Mesa, Lazaro Alonso, Nuno Carvalhais, José Cortés, Zhihan Gao, Nora Linscheid, Mélanie Weynants, and Markus Reichstein. Multi-modal learning for geospatial vegetation forecasting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27788–27799, 2024.

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[4] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. URL https://mlanthology.org/tmlr/2024/oquab2024tmlr-dinov2/.

[5] Christian Requena-Mesa, Vitus Benson, Markus Reichstein, Jakob Runge, and Joachim Denzler. Earthnet2021: A large-scale dataset and challenge for earth surface forecasting as a guided video prediction task. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 1132–1142, 2021. doi: 10.1109/CVPRW53098.2021.00124.

A Expanded Results

A.1 Full headline forecast table

Table 4 expands the main-text headline to all nine encoder views from the Tier-1 re-run. The shared base is $[\mathrm{NDVI}(t), \mathrm{weather}]$; all rows use ridge regression, cube folds, and the plausibility screen.

A.2 Protocol summary

- **Data.** 115 GreenEarthNet minicubes from tile 32UNU, all from 2018.
- **Forecast target.** Cube-mean NDVI at 5, 25, 50, and 100 days.
- **Retained frames and rows.** 1580 retained frames in the rebuilt manifest; 1540 total forecast rows in the Tier-1 run.
- **Plausibility screen.** The shared target builder flags 3 of 1580 retained frames as implausible; removing any forecast row touching one of them drops 8/5/5/0 rows at 5/25/50/100 days, changing row counts from 518/489/450/196 to 510/484/445/196.
- **Cross-validation.** Cube-grouped splits only; claims are reported for `cube`, LOCO, and `spatial_block`. Effective sample size is cubes, not frames.
- **Comparisons.** "X beats Y" is always the paired per-fold difference with a delete-one-fold jackknife interval, not a comparison of overlapping marginal confidence intervals.
- **Controls.** Persistence, proxy climatology, observation-process, horizon-only, and permutation controls travel with the main table. P4 also includes a day-of-year sanity control.

A.3 Paired separability and colour-infrared twins

A.4 Cross-probe handshake

A.5 Extreme-subset finding (Tier-1 screened table)

The main forecast headline is dominated by near-normal rows. On the extreme subset, under the Tier-1 protocol (nested-CV ridge, shared base, screen on):

- At 5 and 25 days, every *frozen network* is negative against persistence in `extreme_low`.
- At 50 and 100 days, networks turn positive on `extreme_low` but remain negative on `extreme_high` (DINOv2 reaches -0.62 at 100 d).
- No frozen network is positive on *both* extreme tails at any horizon.
- Hand-crafted `raw_rgb_only` clears both tails at 5 and 25 d ($+0.12/+0.75$ and $+0.30/+0.38$) and still fails on `extreme_high` at 50–100 d (-0.72, -0.57).

This is why the paper treats sign-versus-magnitude—and tail skill versus pooled R^2—as climate-relevant rather than merely diagnostic: a model can look strong overall and still miss the windows an early-warning trigger would fire on.

A.6 Proxy climatology check (30TVN)

On seasonal tile 30TVN (87 cubes), Stage A proxy weather skill is $+0.192\,[+0.139, +0.245]$ while Stage B leave-year-out skill is $-0.136\,[-0.852, +0.581]$. The point estimates differ by 0.328; the intervals overlap because `crossed` folds clamp to four years. That is *not* evidence the proxy is sound—it is the absence of evidence either way—and no H1-style number is quoted from Stage B.

A.7 Plausibility-screen alignment

P3 applies `frame_plausible` (in-season cube-mean NDVI below 0.15 flagged as cloud). Screening 8 of 518 forecast rows at $\Delta = 5\,\mathrm{d}$ moved persistence from $+0.169$ to $+0.690$. P2 and P4 inherit the same opt-in flag; screened CSVs are written beside the cubes as `p2_screened_results.csv` / `p4_screened_results.csv` so the published unscreened tables remain reproducible.

Model row	5 d	25 d	50 d	100 d
raw_features + weather	0.871	0.743	0.625	0.639
raw_rgb_only + weather	0.853	0.727	0.589	0.556
$[\mathrm{NDVI}(t), \mathrm{weather}]$ only	0.773	0.693	0.522	0.406
imagenet_vit_b16_cir	0.712	0.670	0.524	0.451
satlas_s2_swinb_rgb_cir	0.698	0.647	0.514	0.370
satlas_s2_swinb_mi_rgb	0.640	0.667	0.626	0.570
imagenet_vit_b16	0.656	0.622	0.542	0.421
satlas_s2_swinb_rgb	0.616	0.620	0.559	0.500
dinov2_vitb14	0.622	0.577	0.512	0.550
dinov2_vitb14_cir	0.586	0.551	0.517	0.586
persistence	0.690	0.542	0.124	0.129
proxy climatology	0.026	0.125	0.136	-0.164
observation control	0.014	0.056	0.072	-0.024
permutation null	-0.014	-0.002	-0.007	-0.024
horizon-only control	-0.006	-0.006	-0.007	-0.023

Table 4: Tier-1 P3 headline table from the 115-cube re-run. Effective n in cubes is 115/114/115/94 at 5/25/50/100 days. The multi-image Satlas row is kept for completeness but is not single-image comparable.

Question	Measured answer
Do frozen encoders beat the band-matched baseline?	No. At 5 days all 9 encoder views are separably below `raw_rgb_only` by -0.141 to -0.267; at 25 days 4 of 9 are separably below; at 50 days none is separable either way; at 100 days 3 of 9 are separably below and none above.
Does NIR access rescue the frozen networks?	No. Across RGB/CIR twins, 16 of 48 comparisons are separable and 22 of 48 have the CIR row ahead at all. Gains of $+0.05$ to $+0.14$ appear mainly at 5 and 25 days for two single-image encoders, but never enough to reach the band-matched baseline.

Table 5: Appendix summary of the paired tests that underpin the main claim.

Probe	Key number used in the paper
P2 sign	DINOv2 reaches $\rho = 0.606\ [0.546, 0.665]$; the band-matched `raw_rgb_only` row reaches 0.695, above every network.
P2 magnitude	All correlations are at most 0.12; margins are at most $+0.07$; 3 of 4 encoders flip margin sign across fold modes.
P4 proxy ceiling	At `cell_mean` under HGB, weather explains $0.130\ [0.063, 0.197]$ on `cube` and $0.085\ [0.007, 0.162]$ on LOCO; 17 of 54 weather rows exclude zero at 115 cubes.

Table 6: Numbers from the complementary probes used to interpret the main forecast result.